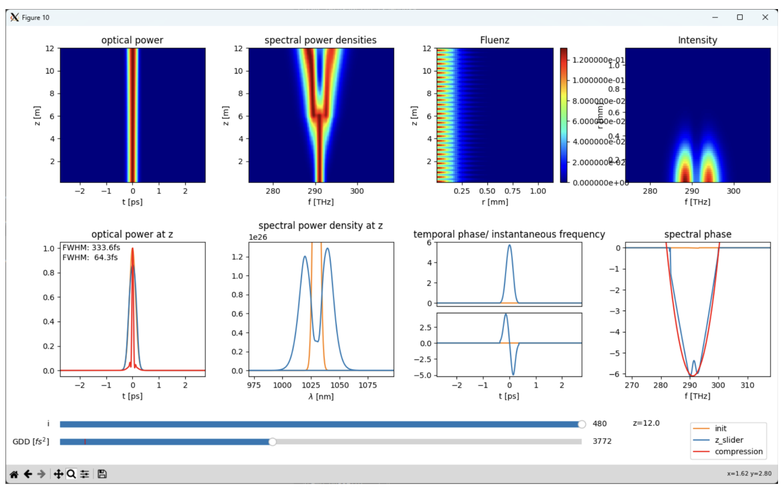
Implementation of a Compact Multipass Cell Pulse Compressor System with a Second-Order Autocorrelator

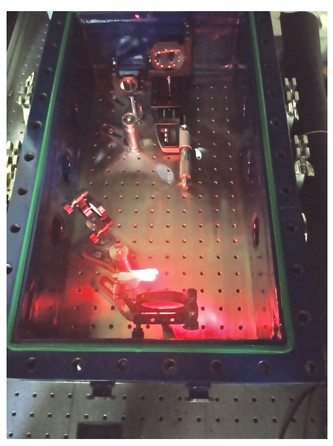
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Multipass Cell

- Multi pass cells can be used for **pulse compression** by allowing a laser pulse to undergo multiple passes within a non-linear medium (in our case Argon gas) to accumulate **Self-Phase Modulation** (SPM), broadening its spectrum.
- A set of dispersive/chirped mirrors are then used to compensate the chirp induced by SPM, temporally compressing the pulse.
- We are setting up a multi pass cell based on [1]



(a) Simulation of multipass cell geometry



(b) Experimental multipass cell setup

Autocorrelation: A theoretical paradigm

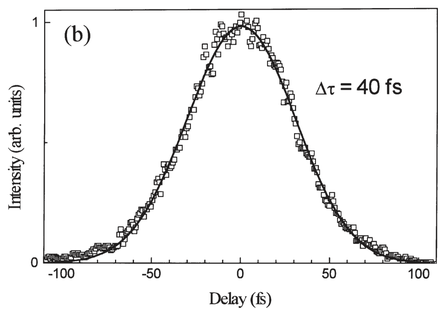
- Converts a temporal delay which is very small for femtosecond pulses into a spatial delay which is easier to create using an **interferometer**
- First order autocorrelation gives coherence length of the laser pulse which cannot be related to pulse width in general. To determine pulse width, we turn to the **second order** autocorrelation function:

$$I_2(\tau) = \int_{-\infty}^{+\infty} |[\mathbf{E}(t) + \mathbf{E}(t-\tau)]^2|^2 dt$$

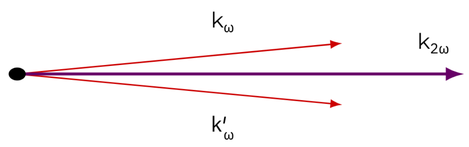
- Expanding this relation with a general pulse profile we get:

$$I_2(\tau) = \int_{-\infty}^{+\infty} \Big| 2E^4(t) + 4E^2(t)E^2(t-\tau) + 4E(t)E(t-\tau)[E^2(t) + E^2(t-\tau)]\cos[\omega\tau + \Phi(t) - \Phi(t-\tau)] + 2E^2(t)E^2(t-\tau)\cos[2(\omega\tau + \Phi(t) - \Phi(t-\tau))]\Big| dt$$

- For intensity autocorrelation, the signal at the peak (zero delay) is 3 times its value at large delays giving an autocorrelation trace as below (background subtracted) [2]:



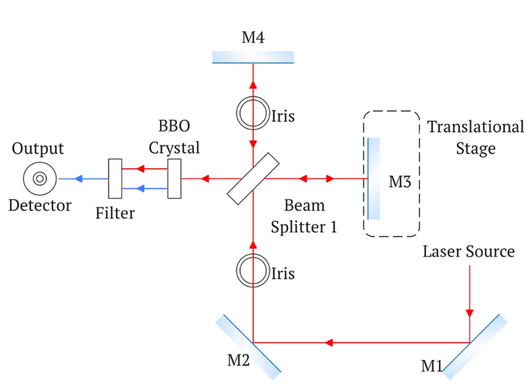
(a) Expected intensity autocorrelation trace.



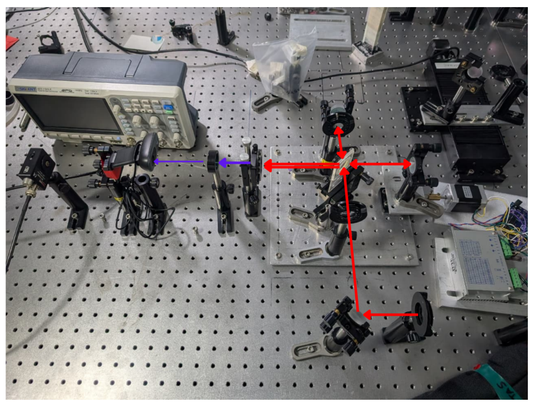
(b) Phase matching in non-collinear SHG geometry.

- The width of the autocorrelation trace can give the width of the pulse.

Experimental Setup



(a) Schematic



(b) Experimental setup

Setup Description

In laboratories with sources of vibrational noise, designing an autocorrelator requires a robust arrangement of optics and high-caliber delay stages to obtain definitive results. We employ a **Michelson Interferometer** for this purpose:

- Using an **800nm Astrella Ultrafast laser**, we first align the beam towards Beam Splitter 1.
- At the beam splitter, we obtain 2 beams, of which one is sent towards the in-house translational stage for adding delay.
- The beams from M2 and M3 are then sent into the SHG crystal at an angle with respect to each other (non-collinear setup). When phase matched, **second harmonics** are obtained at an intermediate angle due to interference.
- The pulse is then passed through a **band stop filter** to remove the fundamental wavelength and into a **photodiode**.
- The peak value of the photodiode signal is our autocorrelation signal.

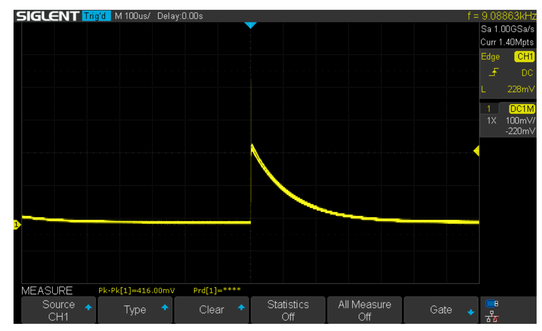
Data Analysis and Outcomes

Results of our experiments require strong proof owing to the noise interference. Fringes also move due to the vibrations in the lab. In order to verify the authenticity of our output signal, we use Signal Processing techniques such as **Fast Fourier Transform** (FFT) and **Curve-fitting** to ameliorate the noisy signal.

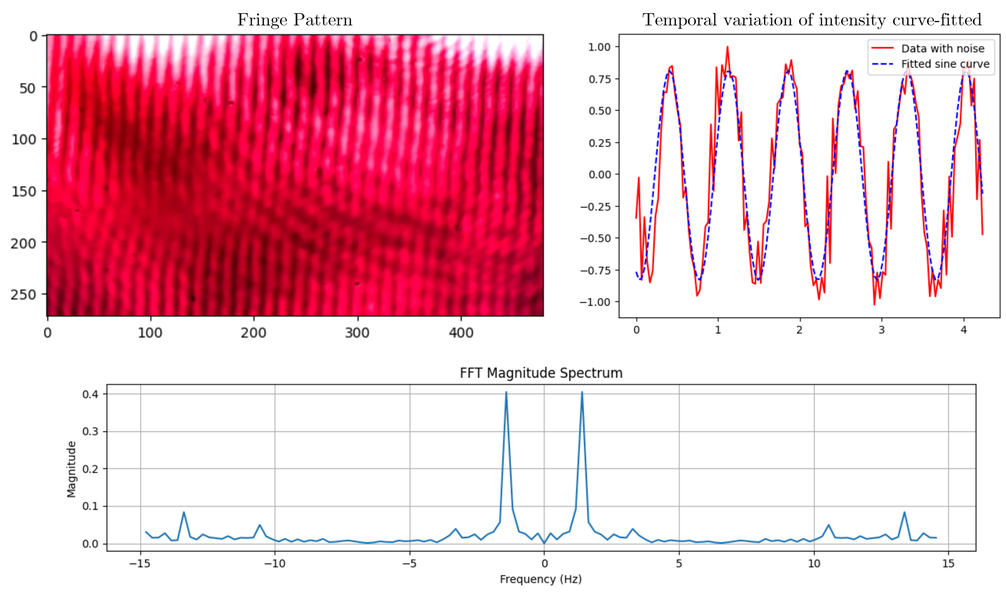
Algorithm 1 Fringe Verification

1: Load video, extract N frames
2: Define Region of Interest
3: **for** $i \leftarrow 1$ **to** N **do**
4: $\quad S[i] \leftarrow \sum_{y \in \text{ROI}} I_i(y)$
5: **end for**
6: Fit $S_{\text{fit}}(t) = A\sin(\omega t + \phi) + B$
7: FFT(S'') to verify frequency

Fringe Verification Algorithm



Photodiode Trace



Extracted interference fringes and fitted sinusoidal envelope.

Future Work

Fine-tuning 2nd and **3rd order Autocorrelation** trace, implementation of the MPC setup to achieve pulse compression for HHG Applications, and improvements in resolution of motorized stage for higher accuracies.

References

References

[1] L. Lavenu, M. Natile, F. Guichard, Y. Zaouter, X. Delen, M. Hanna, E. Mottay, and P. Georges, "Nonlinear pulse compression based on a gas-filled multipass cell," *Opt. Lett.*, vol. 43, pp. 2252–2255, May 2018.

[2] C. Rulliere, *Femtosecond laser pulses : principles and experiments*. Advanced texts in physics, New York: Springer, 2nd ed. ed., 2005.

Acknowledgments

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